PRECALCULUS · UNIT 1: EQUATIONS AND INEQUALITIES

Concert Venue Planner

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students run an outdoor concert. Ticket demand creates quadratic revenue and profit functions, safety tolerances become absolute value inequalities, average cost becomes a rational inequality, and a sound model creates radical equations.

Each solution set is interpreted as a pricing or seating decision.

Standards Addressed

- A-REI.B.4 Solve quadratic equations
- A-CED.A.1–3 Create equations and inequalities; constraints
- A-REI.A.2 Rational and radical equations; extraneous solutions
- Absolute value and polynomial/rational inequalities

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with a concert poster: how should tickets be priced? Part 1.
Day 2	Part 2 rational and polynomial inequalities.
Day 3	Part 3 radical equations and the plan.

Materials

- Student handout and Venue Fact Cards
- Graphing calculators
- Optional: real venue data

How the Three Versions Differ

Level	What changes
CP2	Absolute value tolerances, linear demand, revenue values, and quadratic formula break-even.
CP1	Profit and capacity inequalities, vertex optimization, average-cost rational inequality, and radical equations.
Honors	Vertex form, sponsor changes, profit thresholds, rational inequality sign analysis, and extraneous roots.

Watch For (Common Misconceptions)

- Writing $|L - 100| \geq 6$ for a safe range.
- Including values that make a denominator 0.
- Not checking radical solutions.

Extension Ideas

- Graph profit with technology and verify the vertex.
- Add a second ticket tier (VIP).
- Research decibel safety guidelines.

Answer Key – CP2

Part 1: Tolerances

#	Task	Answer
1	Write an absolute value inequality for the safe sound level L.	$\|L - 100\| \le 6$
2	Solve it. What is the safe range?	$94 \le L \le 106$
3	Write and solve an inequality for the safe stage temperature T.	$\|T - 72\| \le 5 \rightarrow 67 \le T \le 77$
4	The temperature is 79°F. By how much is it outside the safe range?	2°F above 77°F

Part 2: Ticket Demand

#	Task	Answer
5	Simplify the demand equation.	$n = 5{,}000 - 50p$
6	How many fans come if tickets cost $50? $30?	2,500 and 3,500
7	Revenue is $R = p \cdot n = p(5{,}000 - 50p)$. Find R for p = $50.	$50 \times 2{,}500 = \$125{,}000$
8	Solve $5{,}000 - 50p = 4{,}000$ to find the price that exactly fills the venue.	p = $20

Part 3: Break Even

#	Task	Answer
9	Write the cost when the price is p (use n = 5,000 − 50p).	$C = 45{,}000 + 6(5{,}000 - 50p) = 75{,}000 - 300p$
10	Profit $P = -50p^2 + 5{,}300p - 75{,}000$. Use the quadratic formula to find the break-even prices.	$p = \frac{-5{,}300 \pm \sqrt{(28{,}090{,}000 - 15{,}000{,}000)}}{-100} = \frac{5{,}300 \mp 3{,}618}{100} \rightarrow p \approx \$16.82$ or $89.18
11	Recommend a ticket price and explain using your work.	Answers vary; the maximum profit is at p = 53 (vertex), within the break-even range.

Reflection

#	Prompt	Look for
R1	Why is absolute value a good way to describe a tolerance?	It measures distance from a target in either direction.

#	Prompt	Look for
R2	What do the two break-even prices mean?	Between them the concert makes a profit; outside them it loses money.

Answer Key – CP1

Part 1: Revenue & Profit

#	Task	Answer
1	Write revenue R(p) and profit P(p) as polynomials.	$R = 5{,}000p - 50p^2$; $P = -50p^2 + 5{,}300p - 75{,}000$
2	Solve P(p) ≥ 0 to find the prices that make a profit.	$50p^2 - 5{,}300p + 75{,}000 \le 0 \rightarrow 16.82 \le p \le 89.18$
3	Capacity is at most 4,000 fans. Write and solve the inequality for p.	$5{,}000 - 50p \le 4{,}000 \rightarrow p \ge 20$
4	Combine the conditions. What prices are allowed and profitable? What price maximizes profit?	$20 \le p \le 89.18$; vertex p = 53, profit $65,450

Part 2: Average Cost

#	Task	Answer
5	Find A(1,000) and A(3,000).	$51 and $21
6	Solve A(n) ≤ 20 to find how many fans keep the average cost at most $20.	$\frac{45{,}000+6n}{n} \le 20 \rightarrow 45{,}000 \le 14n \rightarrow n \ge 3{,}215$ (n > 0)
7	Why must n > 0 in this inequality? What happens to A(n) as n gets large?	n = 0 makes the denominator 0; A(n) approaches $6.
8	Solve $\lvert L - 100\rvert \le 6$ and $\lvert T - 72\rvert < 5$. Explain the difference in the endpoints.	$94 \le L \le 106$ (included); $67 < T < 77$ (not included)

Part 3: Sound & Safety

#	Task	Answer
9	Solve $112 - 4\sqrt{d} = 100$ to find where the level is 100 dB.	$\sqrt{d} = 3 \rightarrow d = 9$ m
10	Solve $112 - 4\sqrt{d} \le 94$ to find where it is quiet enough for families.	$\sqrt{d} \ge 4.5 \rightarrow d \ge 20.25$ m
11	Write your venue plan: price, expected fans, profit, and the family seating distance.	Answers vary.

Reflection

#	Prompt	Look for
R1	Why must you check the solutions of a radical equation?	Squaring can introduce extraneous solutions.
R2	How do rational inequalities differ from polynomial inequalities?	You must also consider where the denominator is 0, which can never be part of the solution.

Answer Key – Honors

Part 1: Optimal Pricing

#	Task	Answer
1	Find the profit function and the price that maximizes profit by completing the square.	$P = -50(p - 53)^2 + 65{,}450 \rightarrow p = \$53$
2	A sponsor adds \$2 per fan. Find the new profit function and optimal price.	$P = (p - 4)(5{,}000 - 50p) - 45{,}000 = -50p^2 + 5{,}200p - 65{,}000$; vertex at $p = \frac{5{,}200}{100} = \$52$
3	Solve $P(p) > 50{,}000$ for the original model.	$-50(p - 53)^2 + 65{,}450 > 50{,}000 \rightarrow (p - 53)^2 < 309 \rightarrow 35.42 < p < 70.58$
4	Explain why the capacity constraint ($p \geq 20$) does not change the optimal price.	$p = 53$ already gives 2,350 fans ≤ 4,000.

Part 2: Average Cost & Rational Inequalities

#	Task	Answer
5	Solve $A(n) < p$ for the optimal price $p = 53$ and interpret.	$45{,}000 + 6n < 53n \rightarrow n > 957.4 \rightarrow$ at least 958 fans
6	Solve $\frac{n - 2{,}000}{n - 4{,}000} \leq 0$ and give a venue meaning (e.g., comparing attendance to thresholds).	$2{,}000 \leq n < 4{,}000$
7	For which n is A(n) within \$1 of its long-run value \$6? Solve $\lvert A(n) - 6 \rvert \leq 1$.	$\frac{45{,}000}{n} \leq 1 \rightarrow n \geq 45{,}000$ (beyond capacity)
8	Sketch A(n) and describe its asymptotes.	Vertical $n = 0$; horizontal $A = 6$

Part 3: Sound Engineering

#	Task	Answer
9	Solve $112 - 4\sqrt{(d + 5)} = 96$ and check for extraneous solutions.	$\sqrt{(d + 5)} = 4 \rightarrow d = 11$ ✓
10	Solve $\sqrt{(d + 7)} = d - 5$ and check. Interpret one solution.	$d + 7 = d^2 - 10d + 25 \rightarrow d^2 - 11d + 18 = 0 \rightarrow d = 9$ or 2; $d = 2$ gives $\sqrt{9} = -3$ ×; $d = 9$ ✓
11	Design a seating chart: VIP (≥ 100 dB), general (94–100 dB), family (≤ 94 dB). Give each zone's distance range.	VIP $d \leq 9$ m; general $9 \leq d \leq 20.25$ m; family $d \geq 20.25$ m

Reflection

#	Prompt	Look for
R1	How does vertex form reveal the optimal price immediately?	$P = a(p - h)^2 + k$ with $a < 0$ has its maximum k at $p = h$.